Ex-Ante Interaction Regressions

Type 0 vs Type 1 | Oreopoulos replication extension

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1 Research objective and specification

We estimate the immigrant-sounding name penalty in employer callback rates and test whether it differs by three **pre-specified moderators**: language skills, extracurricular activities, and gender. The regression estimates provide a theory-driven counterpart to the causal forest's subgroup treatment-effect estimates; this is **not** a predictive-performance horse race.

Moderator (X)	Variable	Question	Expected interaction
Language skills	`language_skills`	Does listing language proficiency reduce the foreign-name callback penalty?	Positive
Extracurricular activities	`extracurricular_skills`	Do extracurricular signals reduce the foreign-name callback penalty?	Positive
Gender	`female`	Is the foreign-name callback penalty less negative for female than male applicants?	Positive

The positive signs are **ex-ante hypotheses**, not assumed findings. Gender is motivated by gender-differentiated ethnic discrimination, not necessarily by a productivity-signal mechanism.

2 Setup and data preparation

The analysis uses the same cleaned Type 0/1 sample (`df_int1`) as the balance-table and causal-forest scripts. Treatment $D = 1$ denotes a foreign-sounding name (Type 1), and $D = 0$ an English-sounding name (Type 0). Outcome Y is the callback indicator. Inference is clustered by `firmid`.

```
required <- c("dplyr", "estimatr", "knitr")
missing <- setdiff(required, rownames(installed.packages()))
if (length(missing)) install.packages(missing, repos="https://cloud.r-project.org")
library(dplyr)
library(estimatr)
library(knitr)
```

```
if (!file.exists("cleaned data.Rda")) stop("Run cleaned_data.Rmd first, or put cleaned data.Rda in this
loaded <- load("cleaned data.Rda")
stopifnot("df_int1" %in% loaded)
required_cols <- c("D", "callback", "firmid", "fall_data", "type", "language_skills", "extracurricular_s
stopifnot(all(required_cols %in% names(df_int1)))
dat <- df_int1 %>% mutate(
  D = as.numeric(as.character(D)),
  callback = as.numeric(as.character(callback)),
  wave = factor(fall_data)
)
stopifnot(all(na.omit(dat$D) %in% 0:1), all(na.omit(dat$callback) %in% 0:1))
cat("Observations:", nrow(dat), "\n")
```

```
## Observations: 5657
```

```
print(table(dat$D, useNA="ifany"))
```

```
##
##    0    1
## 3026 2631
```

```
print(table(dat$type, dat$D, useNA="ifany"))
```

```
##
##        0    1
##   0 3026    0
##   1    0 2631
```

```
cat("Firms:", dplyr::n_distinct(dat$firmid), "\n")
```

```
## Firms: 3269
```

`same_exp` is excluded from the primary controls because it is systematically missing in wave 2; including it would remove that wave. The common main-effect control set is `language_skills`, `extracurricular_skills`, `female`, `ba_quality`, `ma`, and `exp_highquality`.

3 Baseline average treatment effect

We estimate a wave-adjusted linear probability model:

$$Y_i = \beta_0 + \beta_1 D_i + \delta_{w(i)} + \varepsilon_i.$$

Here β_1 is the wave-adjusted name effect, with firm-clustered CR2 standard errors. We also report a covariate-adjusted version as a robustness specification.

```
ate_wave <- lm_robust(callback ~ D + wave, data=dat, clusters=firmid, se_type="CR2")
ate_adjusted <- lm_robust(callback ~ D + language_skills + extracurricular_skills + female +
                            ba_quality + ma + exp_highquality + wave,
                          data=dat, clusters=firmid, se_type="CR2")


extract_D <- function(model, label) {

  results <- broom::tidy(model, conf.int = TRUE)

  treatment <- results[results$term == "D", ]

  if (nrow(treatment) != 1) {
    stop("Could not find treatment coefficient D")
  }

  data.frame(
    model = label,
    estimate = treatment$estimate,
    se = treatment$std.error,
    ci_low = treatment$conf.low,
    ci_high = treatment$conf.high,
    p_value = treatment$p.value,
    n = nobs(model)
  )
}


kable(bind_rows(extract_D(ate_wave, "Wave-adjusted"),
                extract_D(ate_adjusted, "Wave + covariate adjusted")), digits=4,
      caption="Average name effect (foreign minus English name)")
```

Table 1: Average name effect (foreign minus English name)

model	estimate	se	ci_low	ci_high	p_value	n
Wave-adjusted	-0.0443	0.0069	-0.0579	-0.0307	0	5657
Wave + covariate adjusted	-0.0443	0.0069	-0.0578	-0.0307	0	5657

Note: The standard error (SE) reported above differs from that in Oreopoulos (2011) due to differences in the estimation method. Oreopoulos uses heteroskedasticity-robust (HC1) standard errors, whereas our analysis uses firm-clustered CR2 standard errors (SE = 0.00693) to account for potential within-firm dependence, since multiple resumes were submitted to the same employer. Both methods produce the same ATE estimate

(-0.04434), with the difference in SE arising from the variance estimation method. The table below compares the results using HC1 and CR2, showing that HC1 reproduces Oreopoulos's reported SE of 0.009.

```
#replicating exact Oreopoulos results using HC1 SE

model_HC1 <- lm_robust(
  callback ~ D + wave,
  data = dat,
  se_type = "HC1"
)

#the CR2 SE that we are using instead

model_CR2 <- lm_robust(
  callback ~ D + wave,
  data = dat,
  clusters = firmid,
  se_type = "CR2"
)

# Compare treatment estimates and standard errors
data.frame(
  method = c("HC1", "CR2"),
  estimate = c(coef(model_HC1)["D"],
               coef(model_CR2)["D"]),
  SE = c(model_HC1$std.error["D"],
         model_CR2$std.error["D"])
)
```

```
##   method    estimate          SE
## 1    HC1 -0.04434211 0.008796044
## 2    CR2 -0.04434211 0.006933685
```

4 Three ex-ante interaction regressions

For each moderator X, we estimate:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 (D_i \times X_i) + \boldsymbol{\gamma}' Z_i + \delta_{w(i)} + \varepsilon_i.$$

Z contains the **other five** common controls (the focal moderator is already included as a main effect). β_3 tests whether the name effect differs between $X = 1$ and $X = 0$. The models are fitted separately, using the same full analysis sample where complete.

```
moderators <- c("language_skills", "extracurricular_skills", "female")
controls <- c("language_skills", "extracurricular_skills", "female",
              "ba_quality", "ma", "exp_highquality")
fit_interaction <- function(x) {
  rhs <- c("D", x, paste0("D:", x), setdiff(controls, x), "wave")
  f <- reformulate(rhs, response="callback")
  lm_robust(f, data=dat, clusters=firmid, se_type="CR2")
}
```

```r
models <- setNames(lapply(moderators, fit_interaction), moderators)

interaction_table <- bind_rows(lapply(moderators, function(x) {

  m <- models[[x]]
  term <- paste0("D:", x)

  # Extract regression results with confidence intervals
  results <- broom::tidy(m, conf.int = TRUE)

  # Select the interaction coefficient
  interaction <- results[results$term == term, ]

  if (nrow(interaction) != 1) {
    stop(paste("Interaction term not found:", term))
  }

  data.frame(
    moderator = x,
    interaction = interaction$estimate,
    se = interaction$std.error,
    p_value = interaction$p.value,
    ci_low = interaction$conf.low,
    ci_high = interaction$conf.high,
    n = nobs(m)
  )
}))

kable(interaction_table, digits=4, caption="Ex-ante interaction coefficients (D x X)")
```

Table 2: Ex-ante interaction coefficients (D x X)

moderator	interaction	se	p_value	ci_low	ci_high	n
language_skills	-0.0144	0.0200	0.4715	-0.0535	0.0248	5657
extracurricular_skills	0.0307	0.0183	0.0939	-0.0052	0.0667	5657
female	-0.0366	0.0171	0.0323	-0.0701	-0.0031	5657

5 Regression-implied subgroup treatment effects

For each binary moderator, the model-implied treatment effects are:

$$\widehat{\tau}(X=0) = \widehat{\beta}_1, \qquad \widehat{\tau}(X=1) = \widehat{\beta}_1 + \widehat{\beta}_3.$$

For $X = 1$, the variance must include the covariance between the two coefficient estimates:

$$\widehat{\mathrm{Var}}(\widehat{\beta}_1 + \widehat{\beta}_3) = \widehat{\mathrm{Var}}(\widehat{\beta}_1) + \widehat{\mathrm{Var}}(\widehat{\beta}_3) + 2\widehat{\mathrm{Cov}}(\widehat{\beta}_1, \widehat{\beta}_3).$$

The following code uses each model's firm-clustered covariance matrix and its degrees of freedom to calculate 95% confidence intervals. These subgroup effects, not the raw interaction coefficients, will be compared with `average_treatment_effect(cf, subset = ...)` from the causal forest.

```r
subgroup_effects <- bind_rows(lapply(moderators, function(x) {
  m <- models[[x]]
  b <- coef(m)
  V <- vcov(m)
  interaction <- paste0("D:", x)
  out <- lapply(0:1, function(z) {
    a <- setNames(rep(0, length(b)), names(b))
    a["D"] <- 1
    if (z == 1) a[interaction] <- 1
    estimate <- sum(a * b)
    se <- sqrt(drop(t(a) %*% V %*% a))
    # estimatr stores coefficient-specific degrees of freedom
    df <- min(m$df[match(c("D", interaction), names(b))], na.rm=TRUE)
    crit <- qt(0.975, df=df)
    data.frame(moderator=x, subgroup=paste0("X=", z),
               estimate=estimate, se=se,
               ci_low=estimate-crit*se, ci_high=estimate+crit*se,
               n_subgroup=sum(dat[[x]] == z, na.rm=TRUE))
  })
  bind_rows(out)
}))
kable(subgroup_effects, digits=4,
      caption="Regression-implied subgroup name effects and 95% confidence intervals")
```

Table 3: Regression-implied subgroup name effects and 95% confidence intervals

moderator	subgroup	estimate	se	ci_low	ci_high	n_subgroup
language_skills	X=0	-0.0406	0.0084	-0.0570	-0.0242	4233
language_skills	X=1	-0.0550	0.0167	-0.0878	-0.0221	1424
extracurricular_skills	X=0	-0.0628	0.0130	-0.0883	-0.0372	2240
extracurricular_skills	X=1	-0.0320	0.0101	-0.0518	-0.0123	3417
female	X=0	-0.0257	0.0107	-0.0466	-0.0048	2781
female	X=1	-0.0623	0.0113	-0.0844	-0.0401	2876

6 Interpretation and comparison notes

- A negative effect denotes a callback penalty for foreign-sounding names.
- A positive interaction means the name penalty is **less negative** for $X = 1$ than for $X = 0$.
- The three interaction regressions are separate pre-specified tests; the estimated subgroup effects are model-implied and conditional on the included main-effect controls and wave.
- The causal forest comparison should report doubly robust subgroup ATEs with firm clustering, side by side with these regression estimates and confidence intervals. Neither method observes the true individual CATE.
- Statistical significance of an interaction is not proof of a particular discrimination mechanism. In particular, gender differences need not arise from statistical discrimination.

7 Optional: joint interaction specification

A joint model can be considered as a supplementary analysis, but is not required for the three primary ex-ante tests.

```
joint <- lm_robust(
  callback ~ D * (language_skills + extracurricular_skills + female) +
    ba_quality + ma + exp_highquality + wave,
  data=dat, clusters=firmid, se_type="CR2"
)
summary(joint)
```